

ADITYA VERMA

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks(%)
2021 - 2025	B.Tech in Computer Science and Engineering	Graphic Era University, Dehradun	8.32

EXPERIENCE

Software Engineer, Tata Consultancy Services - Mumbai, Maharashtra November, 2025 - Current

- Designed and built scalable backend systems in Python for a large-scale system modernization initiative at the Reserve Bank of India, architecting the migration of 200+ workflows across 5,000+ institutions from a legacy system to a new distributed architecture.
- Architected bi-directional data processing frameworks handling structured data ingestion and dynamic report generation, powering both real-time query platforms and downstream reporting systems used at enterprise scale.
- Built high-performance distributed processing modules using Python and Apache Spark, writing SQL-based validation and transformation logic to ensure system reliability and data consistency at scale.
- Led development and debugging of large-scale migration systems, identifying and resolving system-level defects to ensure accurate, consistent behavior across legacy-to-new system transitions.
- Collaborated with cross-functional engineering teams in an Agile environment, owning features across the full development lifecycle, from design and implementation to testing and deployment.

PROJECTS

WordSphere: A comprehensive blogging application (August, 2024 - September, 2024) : word-sphere.vercel.app

- Developed a full-stack blogging application using React.js for the frontend and Appwrite as a backend-as-a-service, configuring authentication, database collections, and storage buckets to handle user management and content persistence.
- Implemented complete CRUD functionality for blog posts (creation, editing, deletion) along with advanced content filtering, leveraging Appwrite's REST APIs for secure, scalable data operations.
- Integrated the TinyMCE rich-text editor and styled the application with Tailwind CSS, delivering a polished, accessible writing and reading experience.
- Deployed the application on Vercel, managing the project end-to-end from architecture and backend configuration to production deployment.

Self-Driving Car Agent (September, 2023 - December, 2023) : github.com/addy-301/Self-Driving-Car-Agent-with-RL

- Developed an autonomous driving agent in Unity3D using the ML-Agents toolkit, training and comparing two reinforcement learning algorithms, Proximal Policy Optimization (PPO) and Soft Actor-Critic (SAC), achieving 92% decision-making accuracy and 3.5s average adaptation time to dynamic conditions.
- Trained PPO and SAC models to 700,000 and 1,400,000 steps respectively, tuning training duration to each algorithm's convergence behavior, and benchmarked their learning curves via TensorBoard to identify key differences in sample efficiency and stability.
- Designed a custom simulation track and equipped the vehicle agent with sensors and actuators to enable real-time perception and control, validating end-to-end model performance in a live test environment.

Real-Time Vehicle Tracking System (August, 2023 - September, 2023) : github.com/addy-301/RealTime-Vehicle-Tracking-System-SIH2023

- Engineered a full-stack real-time bus tracking web application using the MERN stack (MongoDB, Express.js, React, Node.js) and Tailwind CSS, progressing from Figma prototypes to production deployment (backend on Azure, frontend on GitHub Pages), achieving 98% real-time location update accuracy through Google Maps API integration.
- Built role-based authentication for drivers (unique ID login) and passengers (route search by origin/destination), paired with a live seat-availability counter so passengers can check bus capacity before boarding.
- Designed safety-first features including a one-tap SOS button that transmits live GPS coordinates to a control room and alerts nearby buses, plus an audible siren system to aid rapid location in emergencies such as accidents.

ACHIEVEMENTS

- **TCS AI Fridays** - Won 1st place among 15 teams at TCS's internal AI hackathon, building a fully functional, well-documented Retrieval-Augmented Generation (RAG) application within a 7-hour time constraint.
- **Intra-University Coding Competition** - Secured 3rd rank in a coding contest hosted by Graphic Era University.
- **TechWizard** - Qualified for Round 2 (twice) of TechWizard, an inter-university coding contest hosted by Graphic Era University.
- **INFathon** - Qualified for Round 2 of INFathon, a coding contest organized by Informatica.

SKILLS

Programming Languages	Python, Java, JavaScript, HTML, CSS, C++
Web Development	React.js, Node.js, Express.js, REST APIs, Tailwind CSS
Database & Backend	MySQL, MongoDB, Apache Hive, Mongoose, Appwrite
AI / Machine Learning	Generative AI, Retrieval-Augmented Generation (RAG), Agentic AI, Reinforcement Learning, Unity ML-Agents
Cloud & Tools	Microsoft Azure, Google Cloud Platform (GCP), Git, GitHub, Docker, Vercel
Core Concepts	SDLC, Agile, Object-Oriented Programming (OOP), Data Structures and Algorithms

CERTIFICATIONS

- **Salesforce Agentblazer Innovator - Salesforce**
- **AWS GenAI Practitioner - Amazon Web Services (AWS)**
- **Introduction to Generative AI - Google Cloud**
- **Cloud Computing Foundations: Cloud Computing Fundamentals - Google Cloud**